

Bhavin K. Moriya

Data Scientist | Machine Learning | Privacy-Preserving Analytics

EU Blue Card Holder (Germany)

bhavinmoriya58@gmail.com

Google Scholar GitHub LinkedIn Kaggle YouTube RoutingApp StreamlitApps



Experience

Data Scientist / Research Assistant	Hochschule Esslingen, Germany	Nov 2023 – Dec 2025
• AnoMoB Project: Developed first-of-its-kind encrypted comparison operations and non-linear function evaluations using OpenFHE (C++) and TFHE-rs (Rust). • Sustainability Analysis: Leveraged SUMO to simulate movement data and conducted comparative analyses of travel modalities focusing on CO2 emissions and optimal route duration. • Performance Benchmarking: Co-authored breakthrough research comparing runtime and memory consumption of CKKS and TFHE schemes, published in Cryptology ePrint Archive (2025). • Data Engineering: Built production-quality software tools for statistical analysis of mobility data using Python, polars, and geopandas.		
Assistant Professor	Univ. of Viçosa, Brazil	Nov 2015 – Nov 2023
• Taught undergraduate and master's level mathematics; supervised 3 dissertations. • Collaborated with international researchers, resulting in 16+ peer-reviewed publications.		

Postdoctoral Fellowships

- IMSc (2011–2013), HRI (2013–2014), CMI (2014), Univ. of Brasília (2014–2015)

Education

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| • Ph.D. in Mathematics , Harish Chandra Research Institute
Thesis: Some Zero Sum Problems in Combinatorial Number Theory | 2011 |
| • M.Sc. & B.Sc. in Mathematics , The M.S. University of Baroda | 2003 – 2005 |

Skills

Programming: Python (Proficient), C++, Rust, SQL, $\LaTeX$, Go, Julia

Data & Simulation: polars, geopandas, SUMO, Scikit-learn, Docker

Workflow: LLM-assisted development (GitHub Copilot), Git, Agile Research, Streamlit

Languages: English (Fluent), German (**B1/Professional Communication**), Portuguese (Fluent/Professional), Hindi (Native), Gujarati (Native)

Selected Projects (GitHub)

- **Vascular Surrogate Models** – Accelerating Intracranial Aneurysm CFD.
- **AudioMNIST** – A spoken digit recognition system.

Publications & Awards

- (2025) **A Performance Comparison of Homomorphic Encryption Schemes CKKS and TFHE**, Cryptology ePrint Archive.
- 16 peer-reviewed publications in combinatorics and cryptography.
- Recipient of multiple international grants (ICM, CIMPA-ICTP, FAPEMIG).